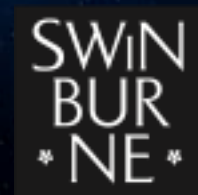


Discovering the Origins of Gravitational-Wave Sources

GR24 and Amaldi16 (2025)

Ashwathi Nair

Primary Supervisor : Dr. Simon Stevenson
Secondary Supervisor : Prof. Jarrod Hurley



Centre for
Astrophysics and
Supercomputing

Discovering the Origins of Gravitational-Wave Sources

Run	Dates	Duration (months)	Detections
O1	September 18, 2015 to January 12, 2016	3	3
O2	November 30, 2016 to August 25, 2017	9	8
O3	April 1, 2019 to March 26, 2020	11	79
			Candidates
O4	May 24, 2023 to March 19, 2025	23	200

O1 to O3 included about 23 months-worth of observing time between 2015 and 2020. In that time, 90 GW detections were made. In the first 23 months of O4, 200 GW candidates have been logged. O4 is currently scheduled to end in November, 2025.

Why care about double neutron stars?



Supernova explosions (SNe)

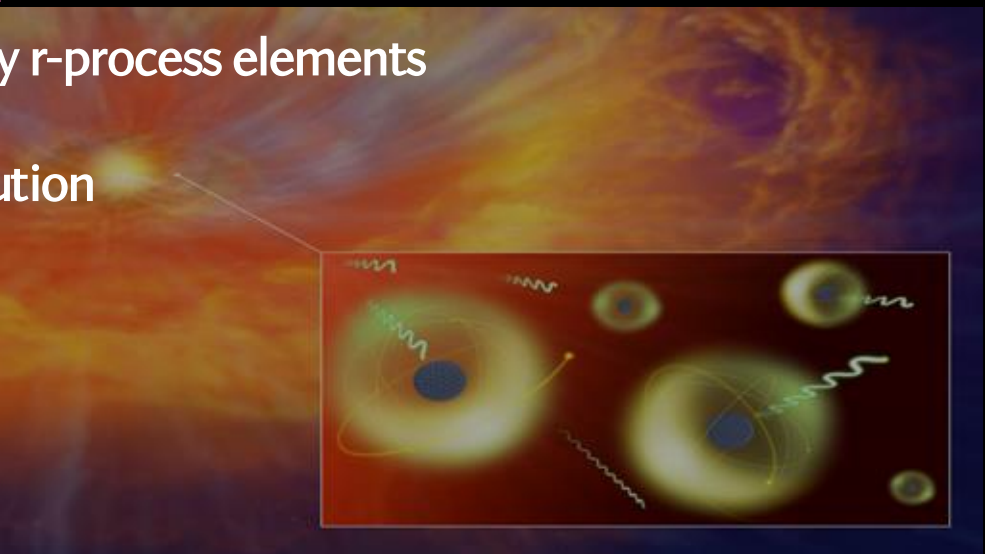
Gamma ray bursts (GRB) and Kilonovae



Formation of heavy r-process elements



Binary stellar evolution



Why care about double neutron stars?

Supernova explosions (SNe)



Gamma ray bursts (GRB)

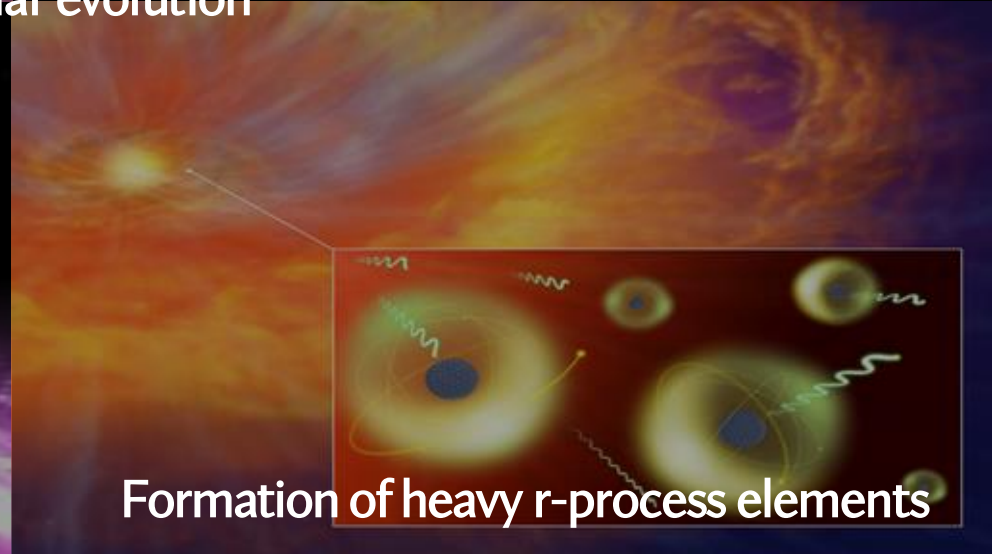


Binary stellar evolution



Kilonovae

Formation of heavy r-process elements



What do we know about Double Neutron Star (DNS) systems so far ?

Detected by radio telescopes and gravitational-wave detectors

Radio Astronomy

~ 26 Galactic DNS in the field with total masses in the range $2.50\text{--}2.89M_{\odot}$

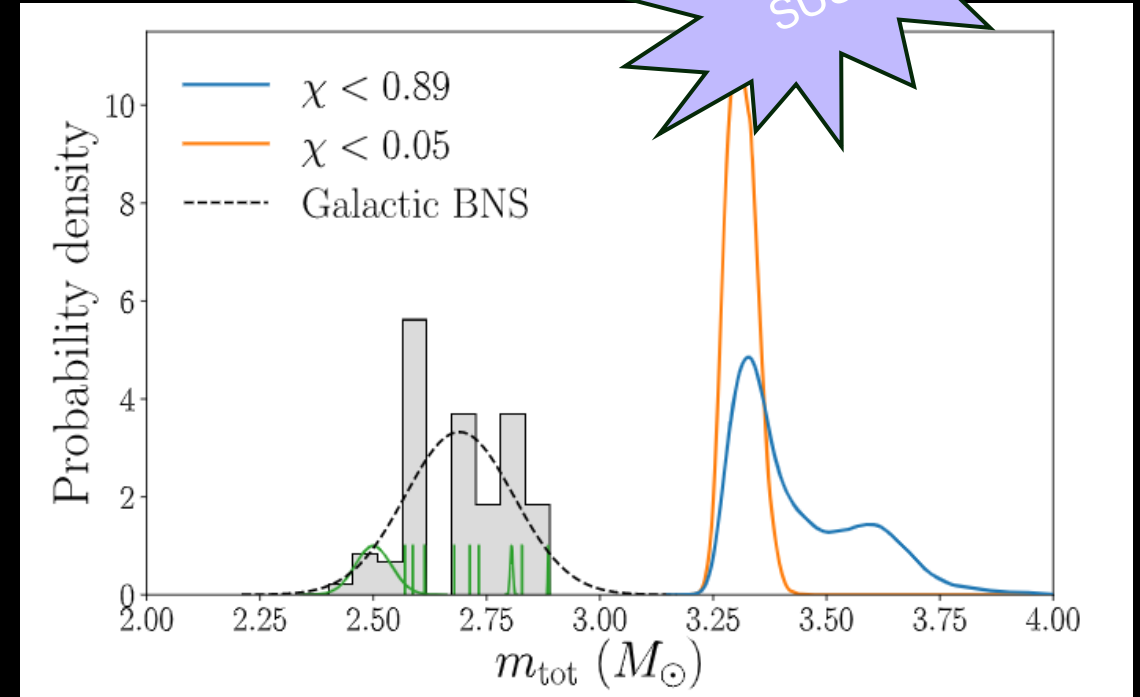
Gravitational Waves

GW170817

Total mass: $2.87^{+0.47}_{-0.09} M_{\odot}$

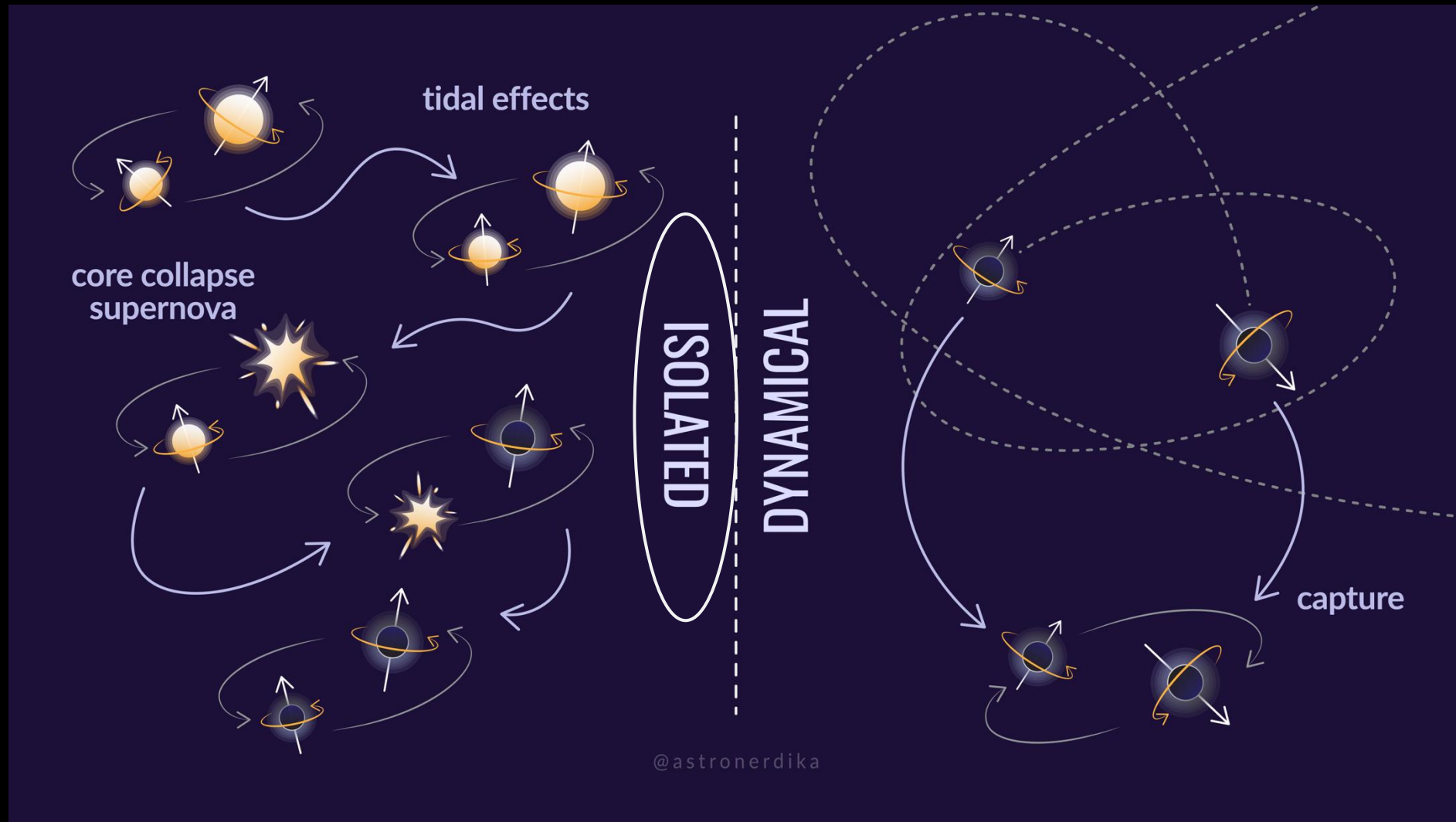
GW190425

Total mass: $3.4^{+0.3}_{-0.1} M_{\odot}$

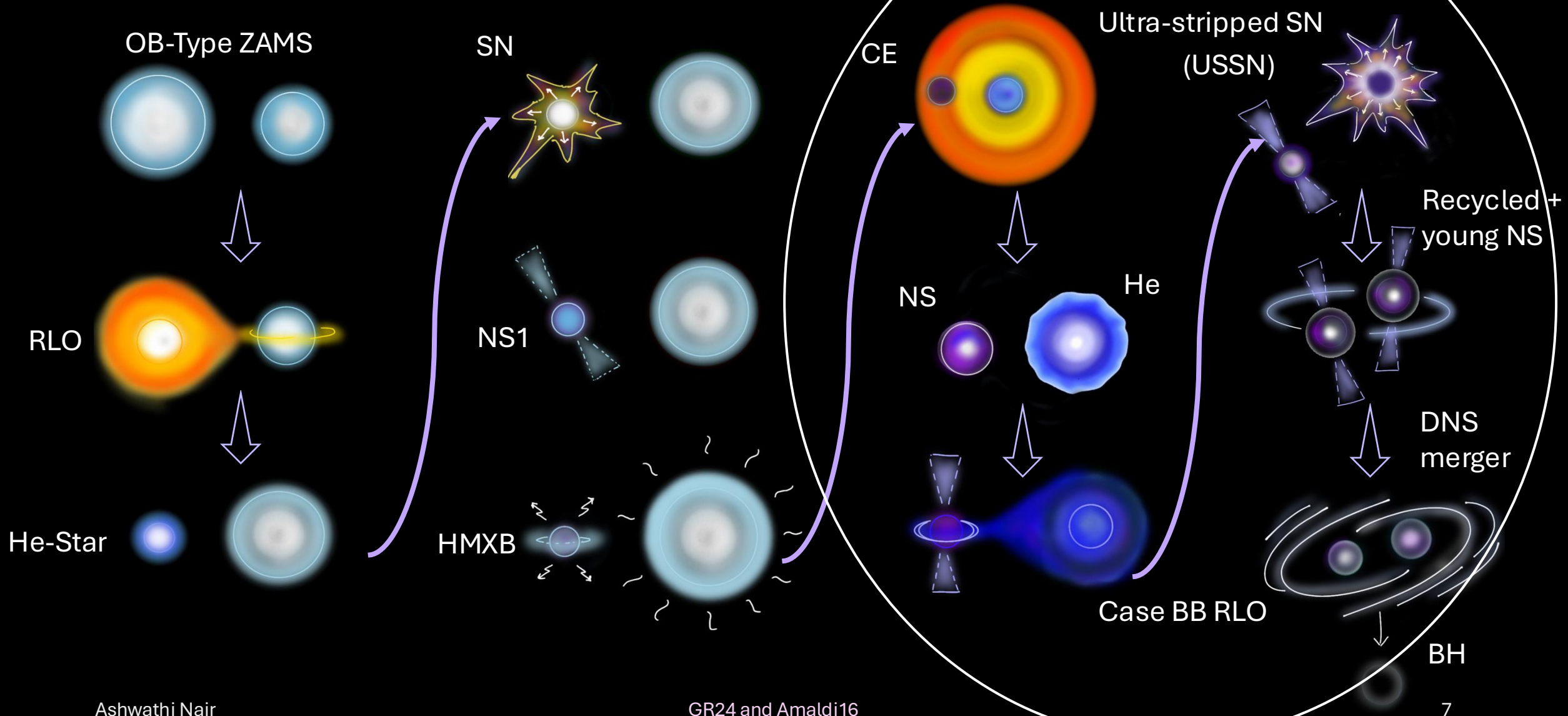


Abbot et al. 2020

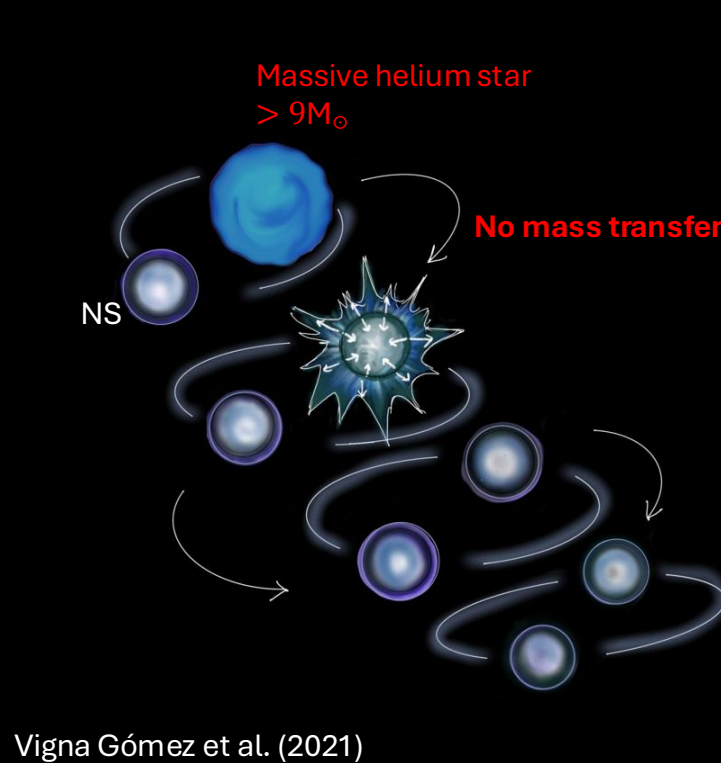
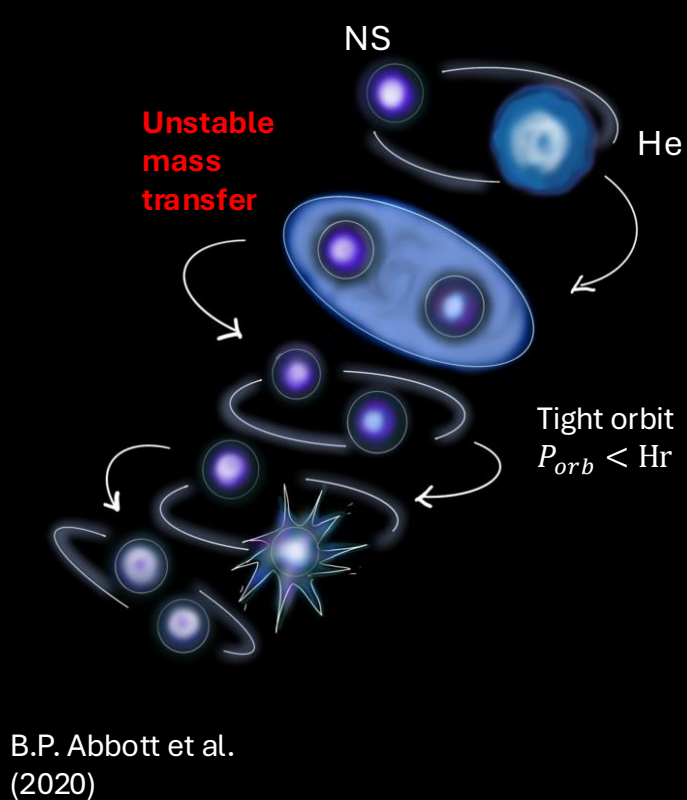
Formation Channels of Compact Binary Mergers



Standard Double Neutron Star Formation Pathway – T. Tauris et al. 2017



Alternate Proposed Formation Pathways for Massive Double Neutron Star



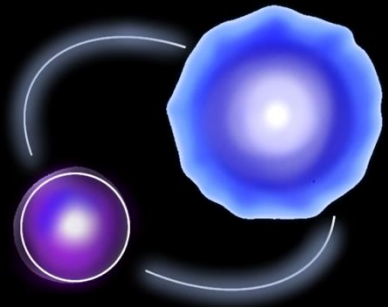
Other uncertainties:

- **Formation at lower metallicities** - Giacobbo & Mapelli (2018)
- **Super-Eddington accretion** - Zhang et al. (2023)
- **Massive first-born NS** - Qin et al (2024)

No self-consistent unbiased formation pathway for heavy double neutron stars

Project I, Part A

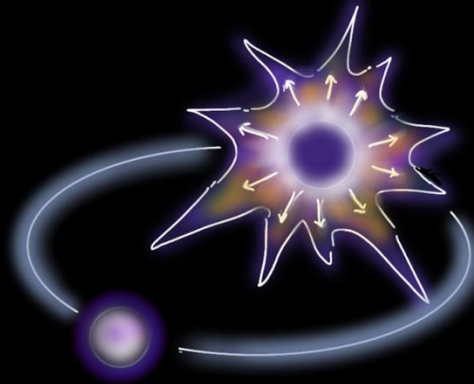
Framework Created for the Formation Pathway



ZAMS He (at solar metallicity)-
 $1.4 M_{\odot}$ NS binary evolution

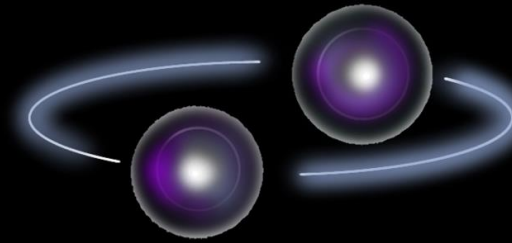
MESA

Paxton et al. 2011



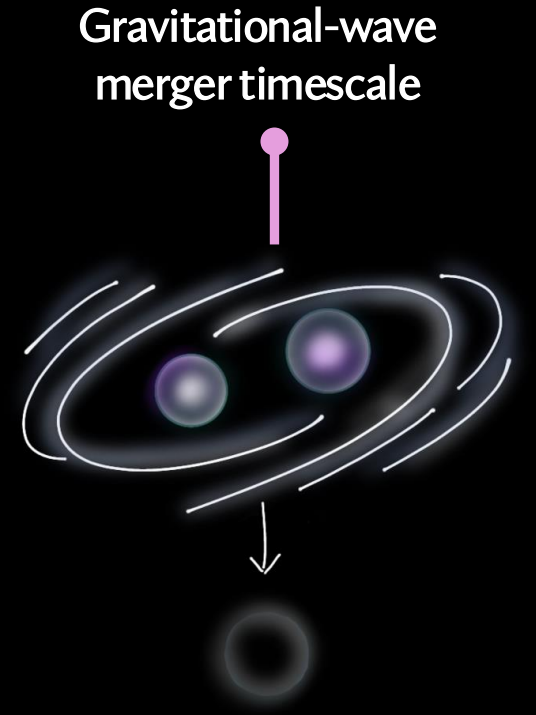
Incorporating supernova and
kick prescription

Mandel & Müller 2020,
Kapil et al. 2023



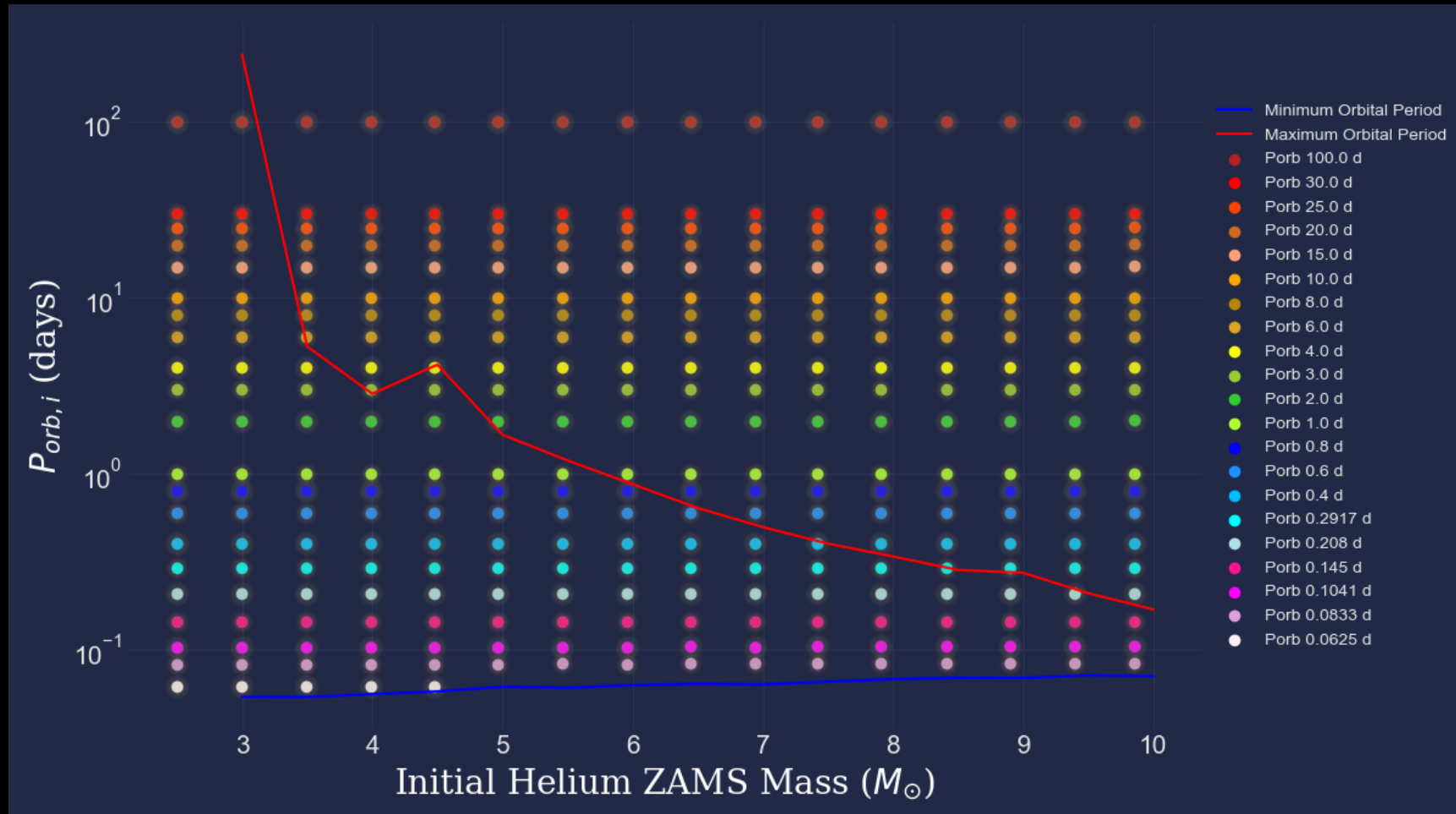
Post-Supernova effects on
orbital dynamics

Brandt &
Podsiadlowski 1995



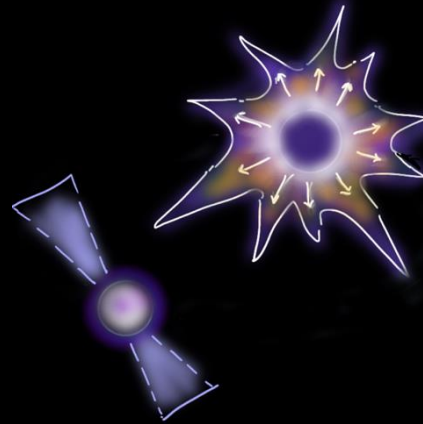
Peters 1964b
Mandel (2021b)

Grid of Models Based On Initial He ZAMS (solar metallicity) Mass + $1.4 M_{\odot}$ NS and Initial Orbital Period

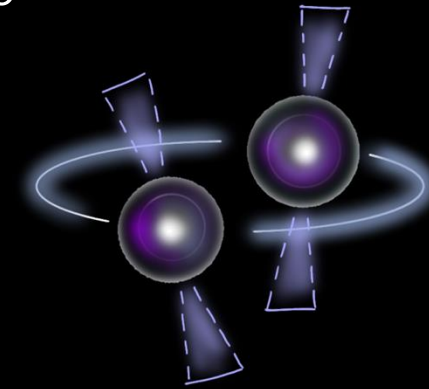


Supernova explosion

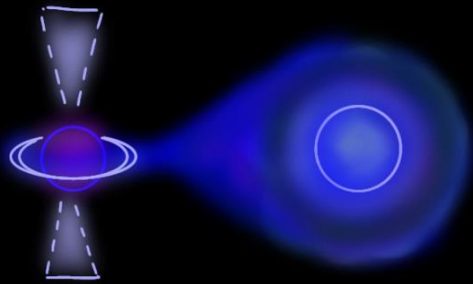
Ultra-stripped SN, Electron
Capture SN, Core-collapse SN



Recycled + young newborn
NS



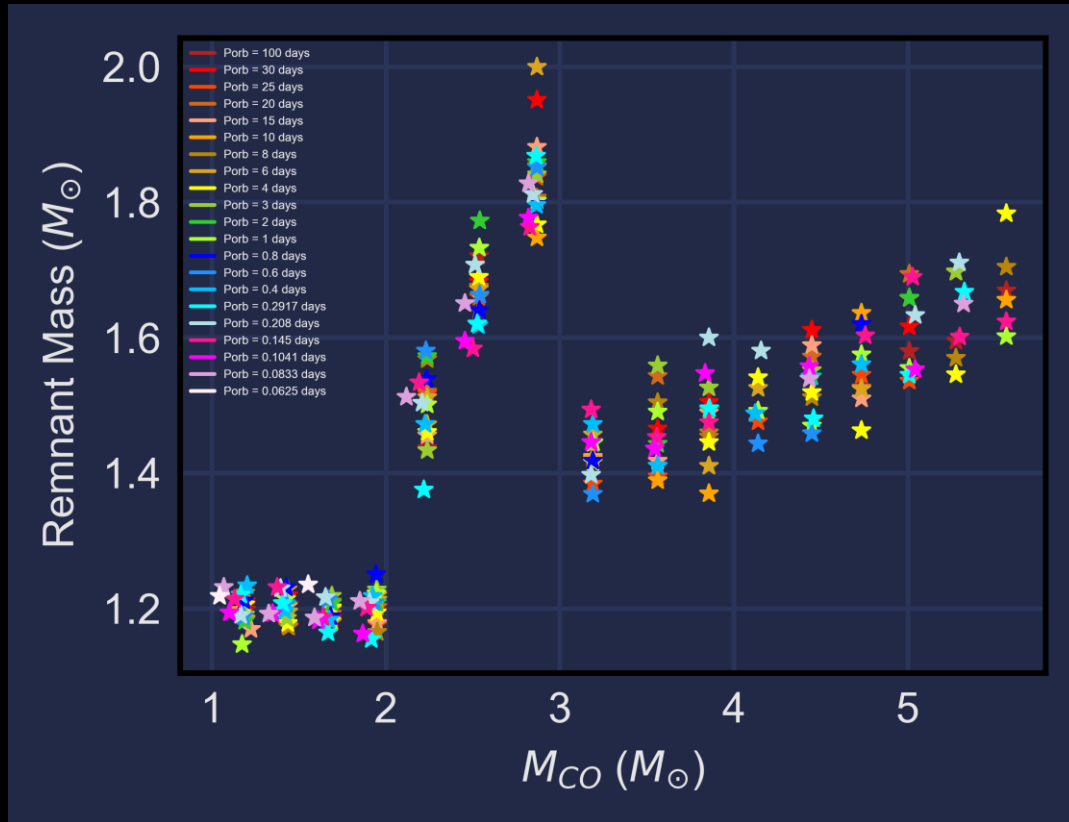
Mass transfer



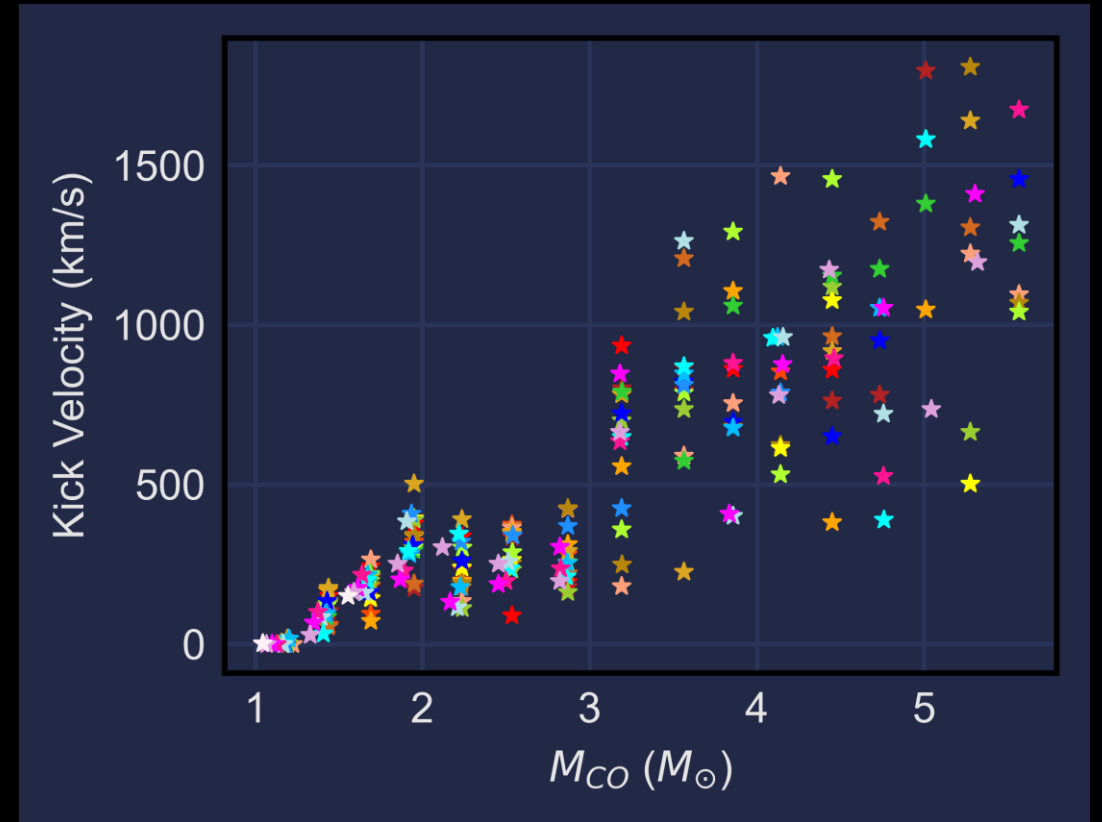
Incorporating supernova and
kick prescription

Mandel & Müller 2020,
Kapil et al. 2023

Standard Supernovae and Kick Prescription (Standard Model)



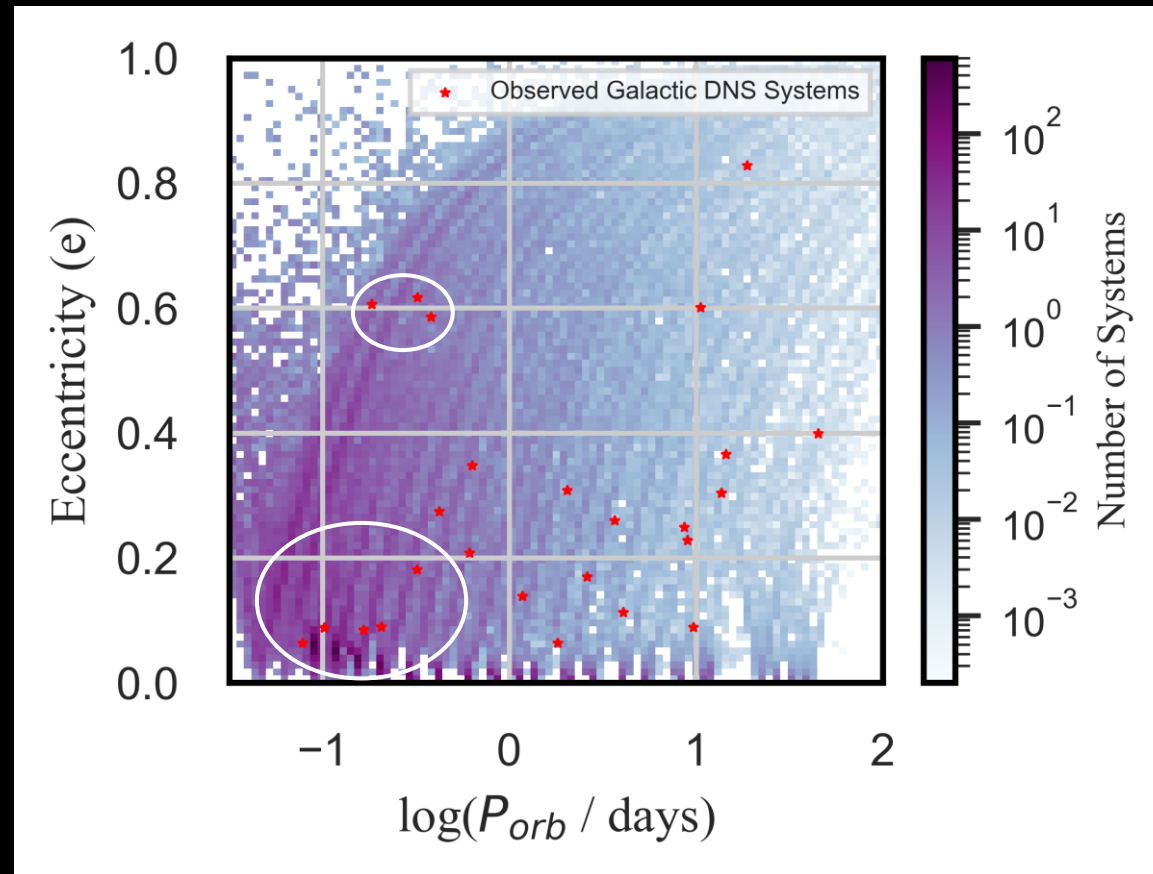
Remnant masses (NS2) as a function of CO core mass



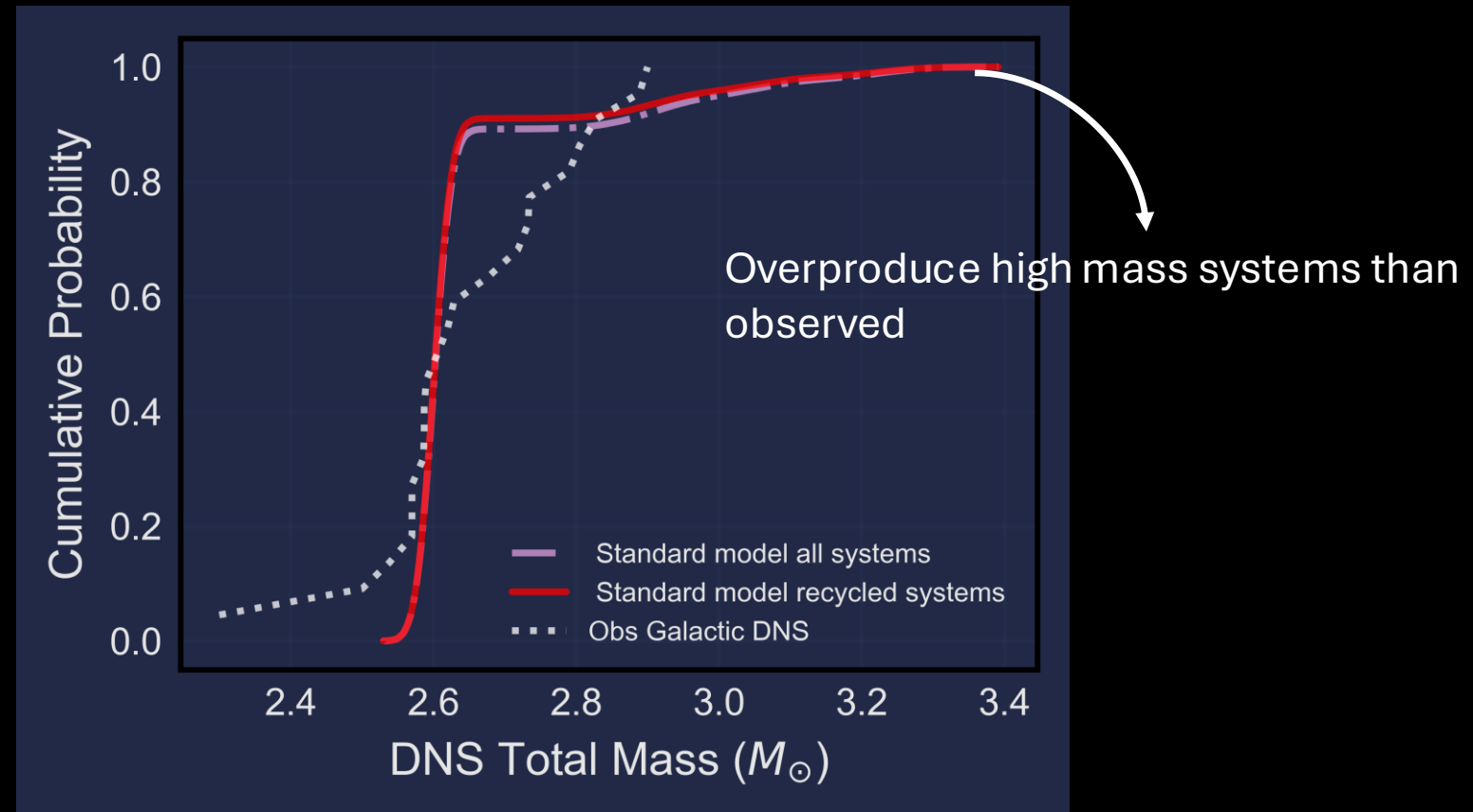
Kick velocity for the corresponding NS2 as a function of CO core mass

Combined P_{orb} -e Distribution of all models

Standard Model



DNS Total Mass Distribution Standard Model



Issues with the Standard Prescription – Mandel & Mueller 2021

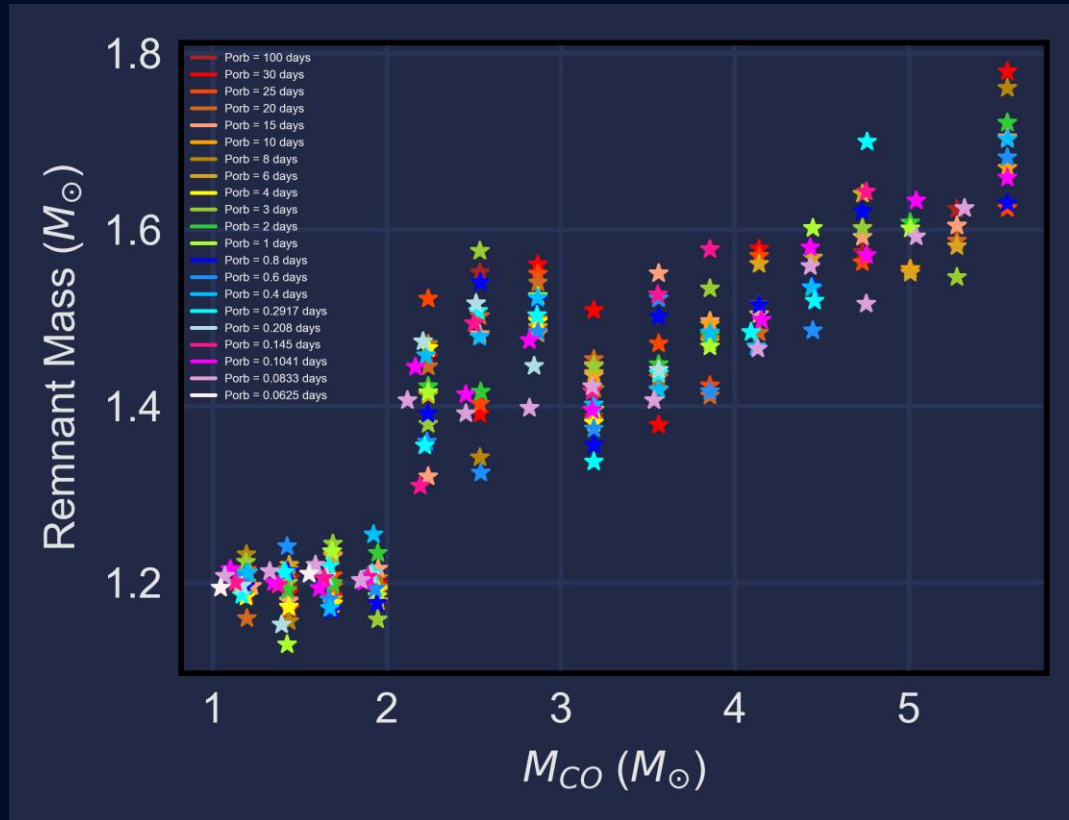
- Overpredicts orbital eccentricities of the systems
- Overpredicts DNS total mass compared to what we observe at solar metallicity

Modifications to the Standard Model

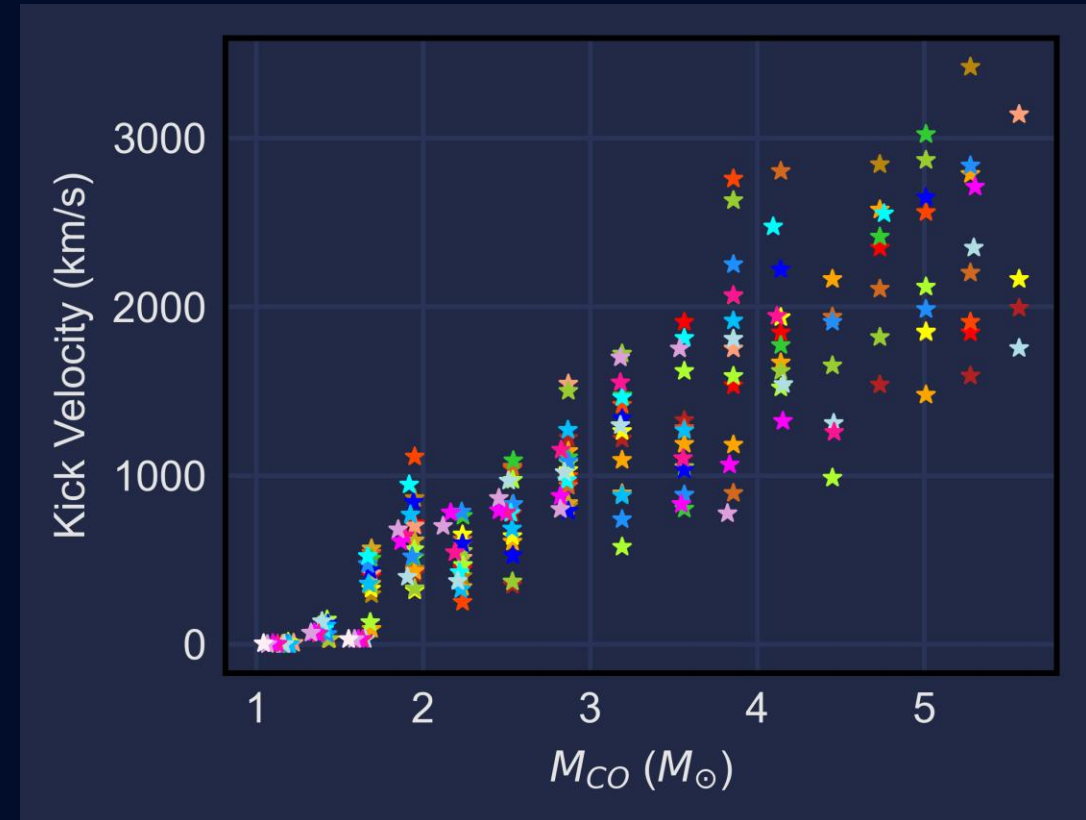
1. Different remnant mass and CO core mass relation
2. Incorporate Ultra-stripped and Electron capture supernova conditions
3. Kick slow merging systems to high velocities

Modified Model

Change this plot



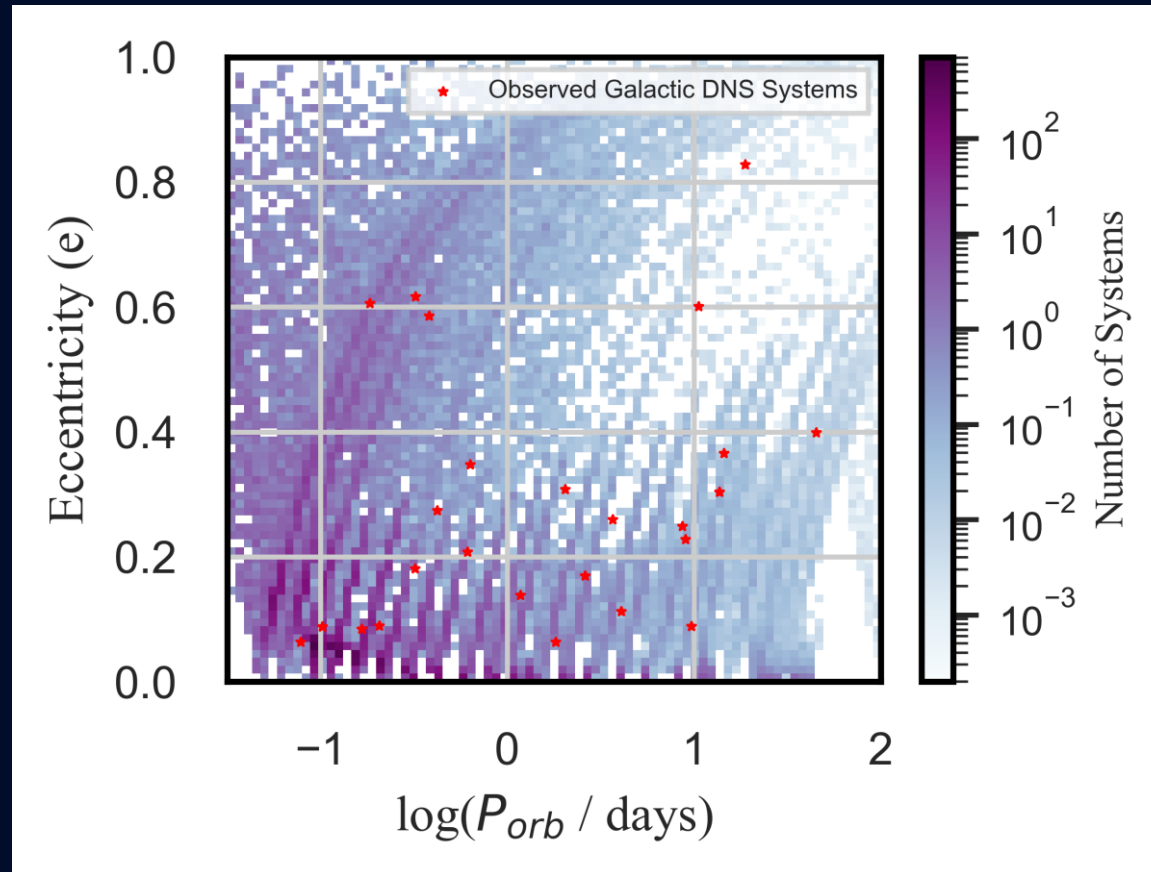
Remnant masses (NS2) as a function of CO core mass



Kick velocity for the corresponding NS2 as a function of CO core mass

Combined Porb-e Distribution of all models

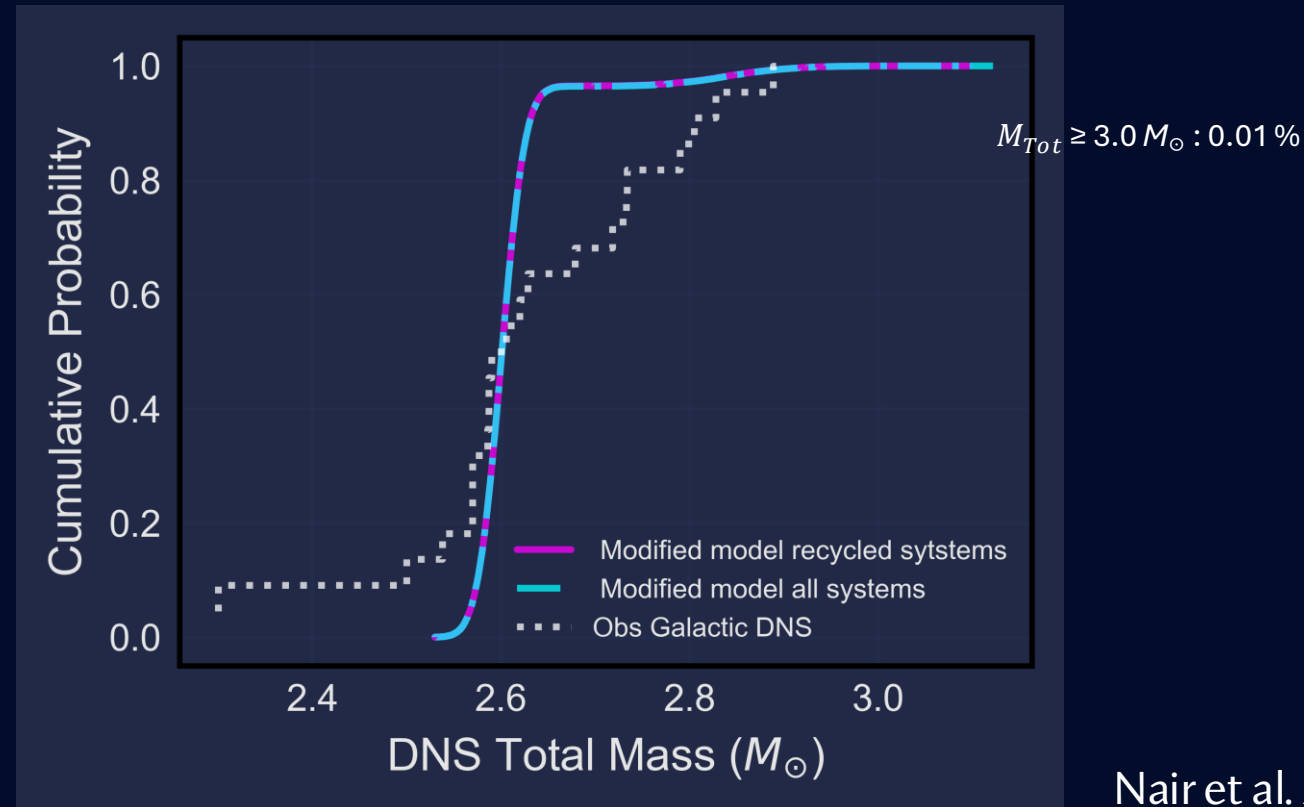
Modified Model



Nair et al. 2025 (Published in MNRAS)

DNS Total Mass Distribution

Modified Model

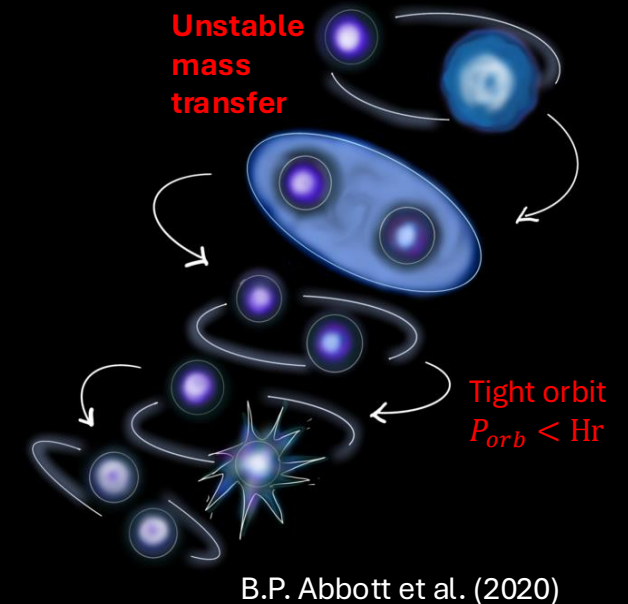
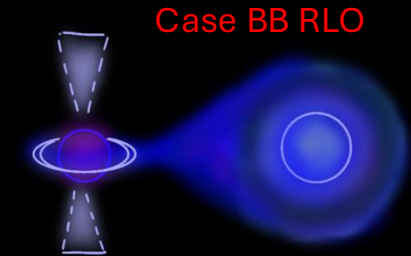
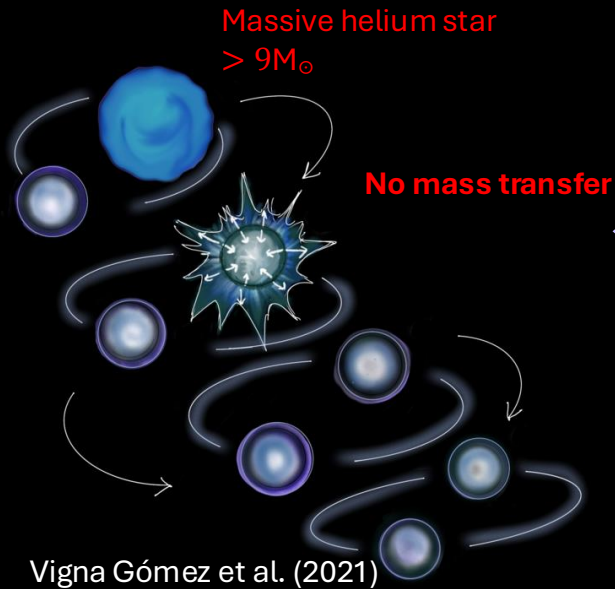


Nair et al. 2025 (Published in MNRAS)

GW190425 Progenitors

Modified: (Out of all the systems $\geq 3 M_{\odot}$)

- 82% of systems - formed via Case BB RLO MT
- 18% of systems - formed via massive helium stars without recycling the first-born NS
- Rules out the formation pathway via unstable mass transfer



Fewer DNS systems $\geq 3 M_{\odot}$

Project I, Part B

Explore

- *Low metallicities*
- *Massive first-born NS*
- *Super-Eddington accretion*

Conclusions



Models calibrated to match the Galactic DNS population



Roughly reproduces the observed Galactic DNS distribution



Used models to explore **GW190425 progenitors**



Massive systems (like GW190425) formed at **solar metallicity** are **rare**



Unstable Case BB MT pathway does not yield GW190425-like binaries



Model limitations: Can't explain high GW190425 rates



Next steps: Explore low-Z models, massive NS primaries & super-Eddington accretion

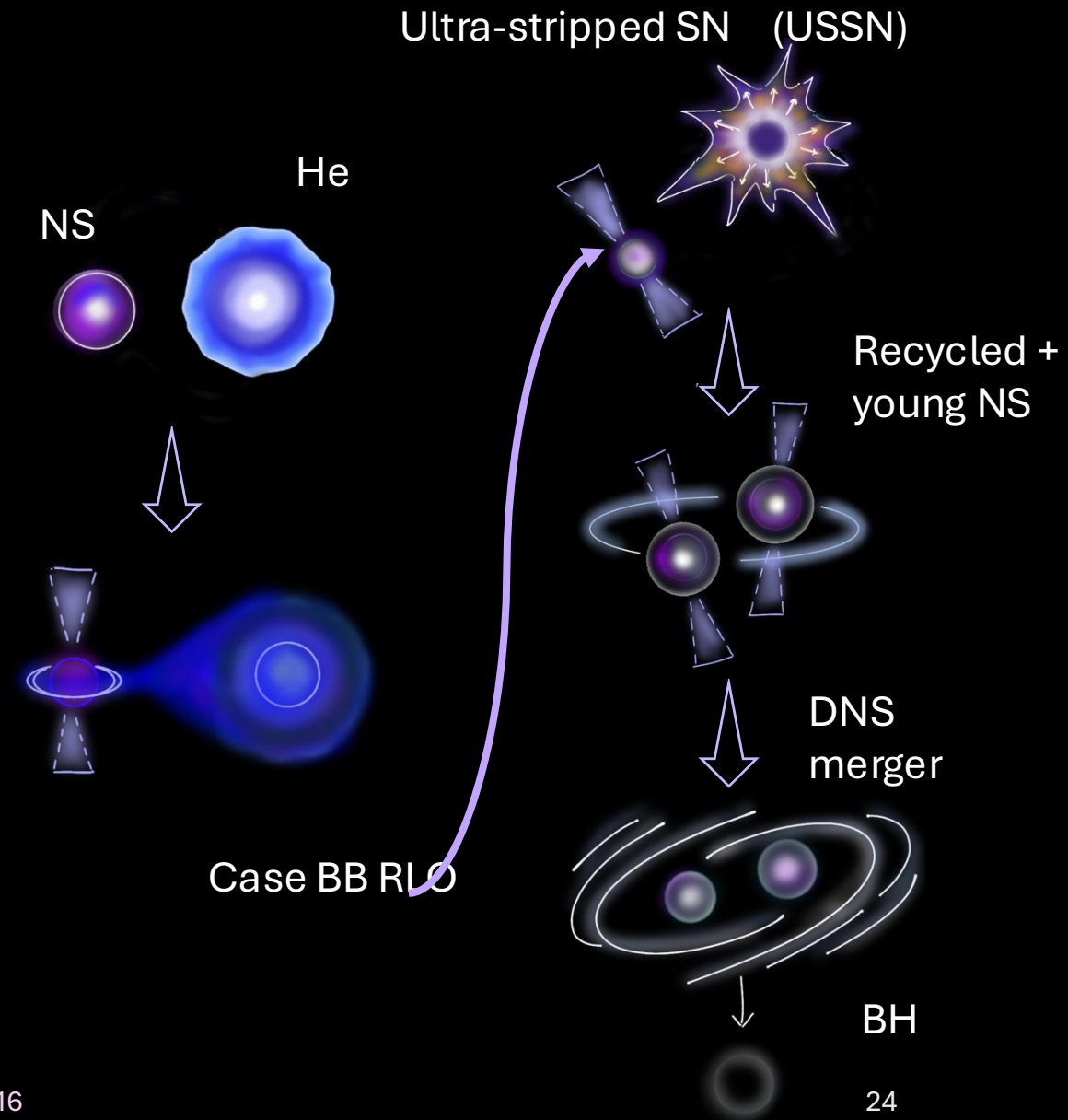


Hoping O4 run detects at least one DNS

EXTRA SLIDES

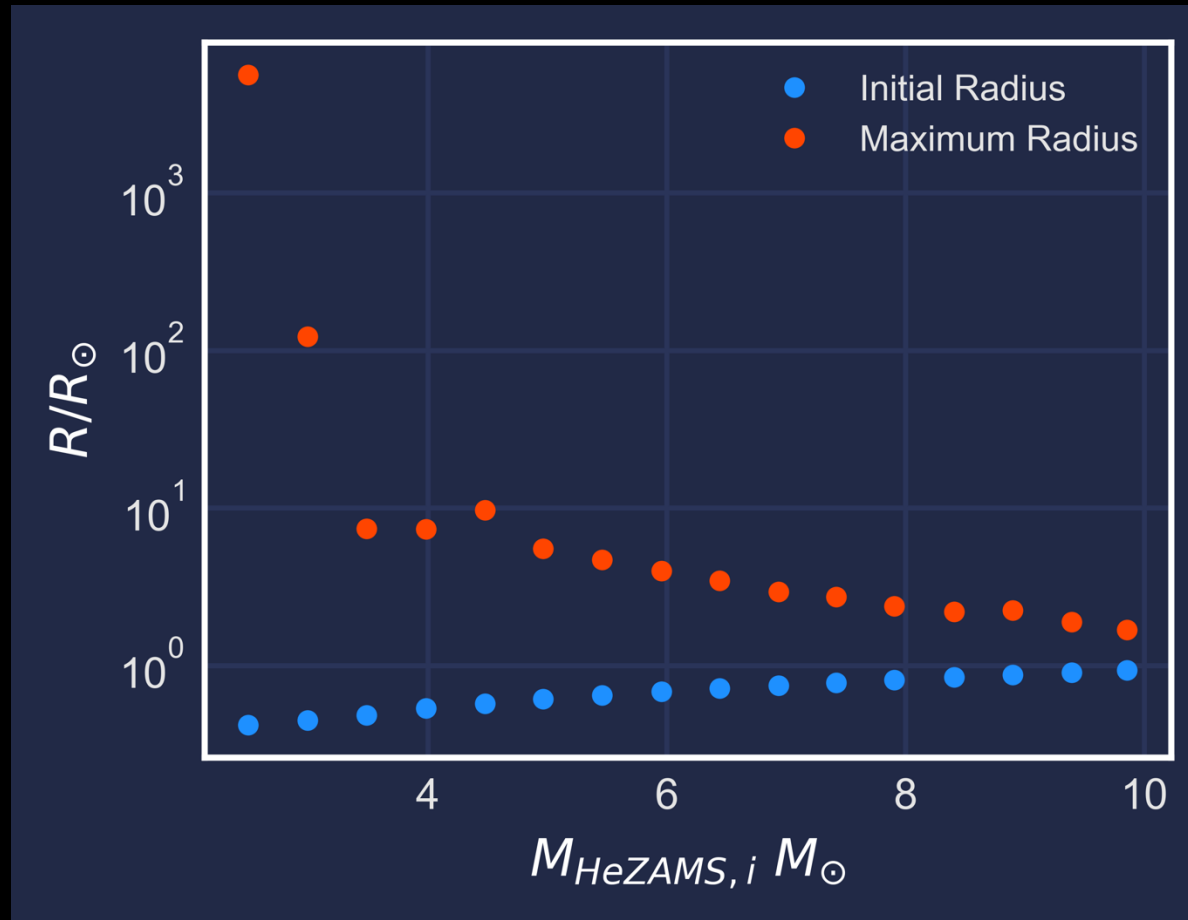
The evolutionary phases that decides the fate of DNSs

Preferred for modeling DNS formation pathways



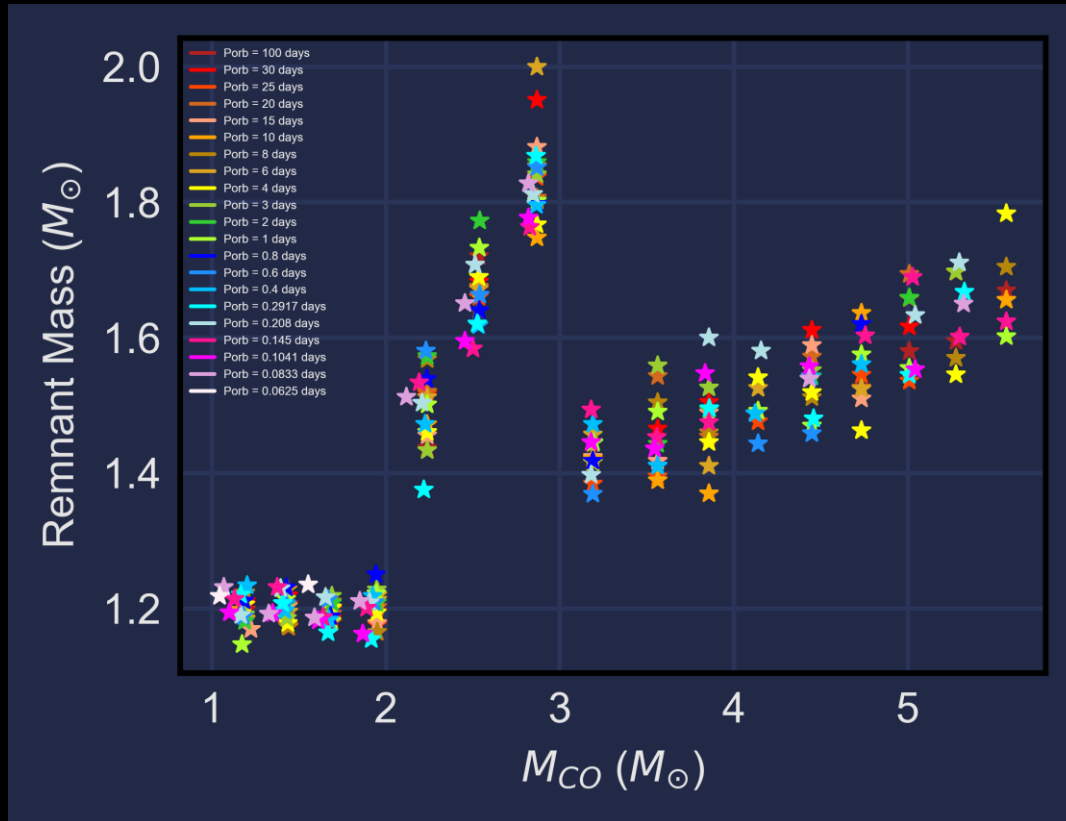
Initial and Maximum Radius as a Function of Initial He-ZAMS Mass at Solar Metallicity

(Recap)

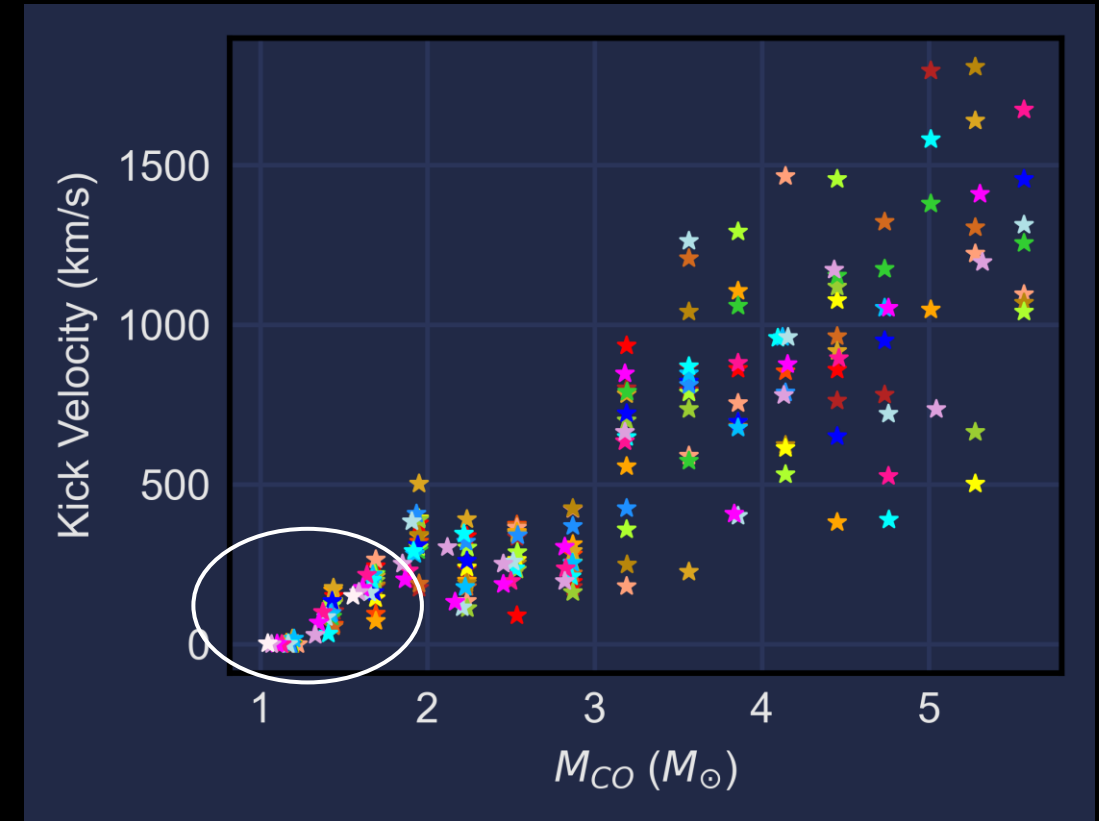


ZAMS :
Zero Age Main Sequence
(aka newly formed stars)

Standard Supernovae and Kick Prescription (Standard Model)



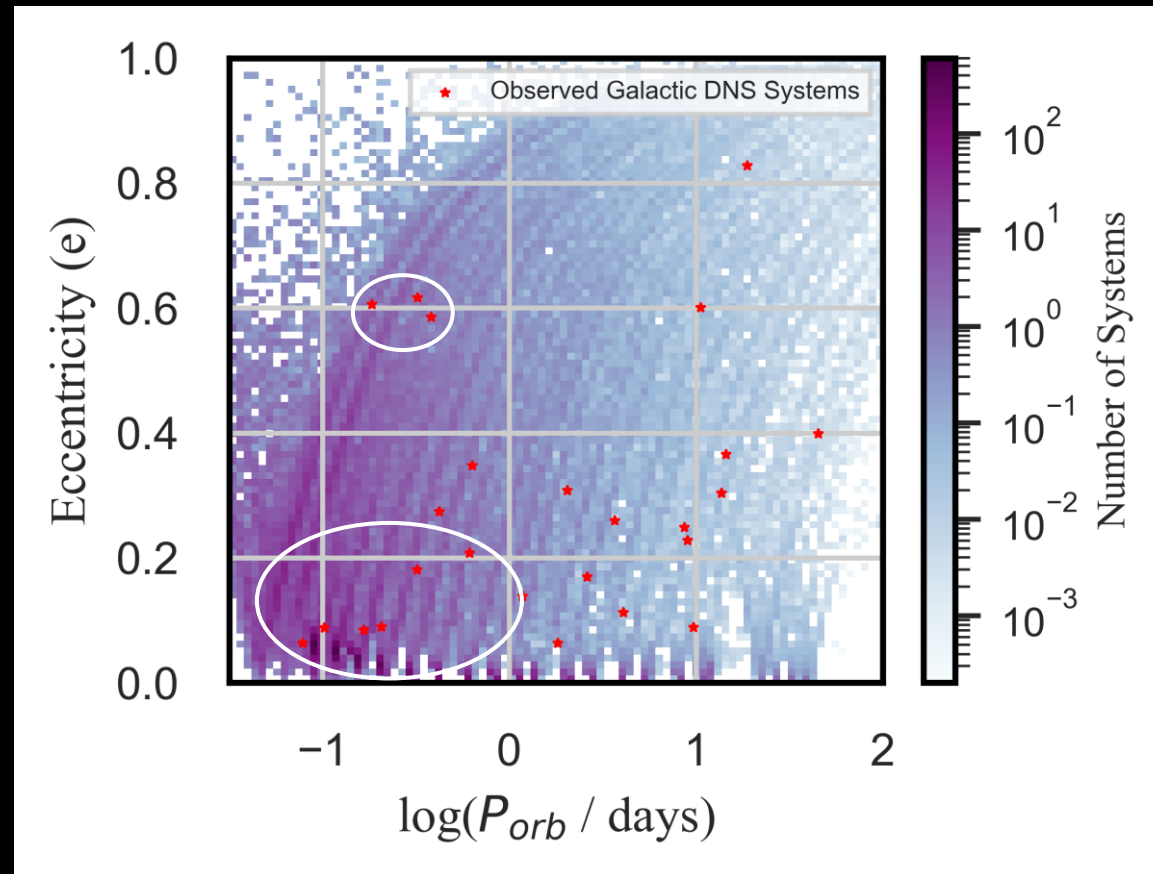
Remnant masses (NS2) as a function of CO core mass



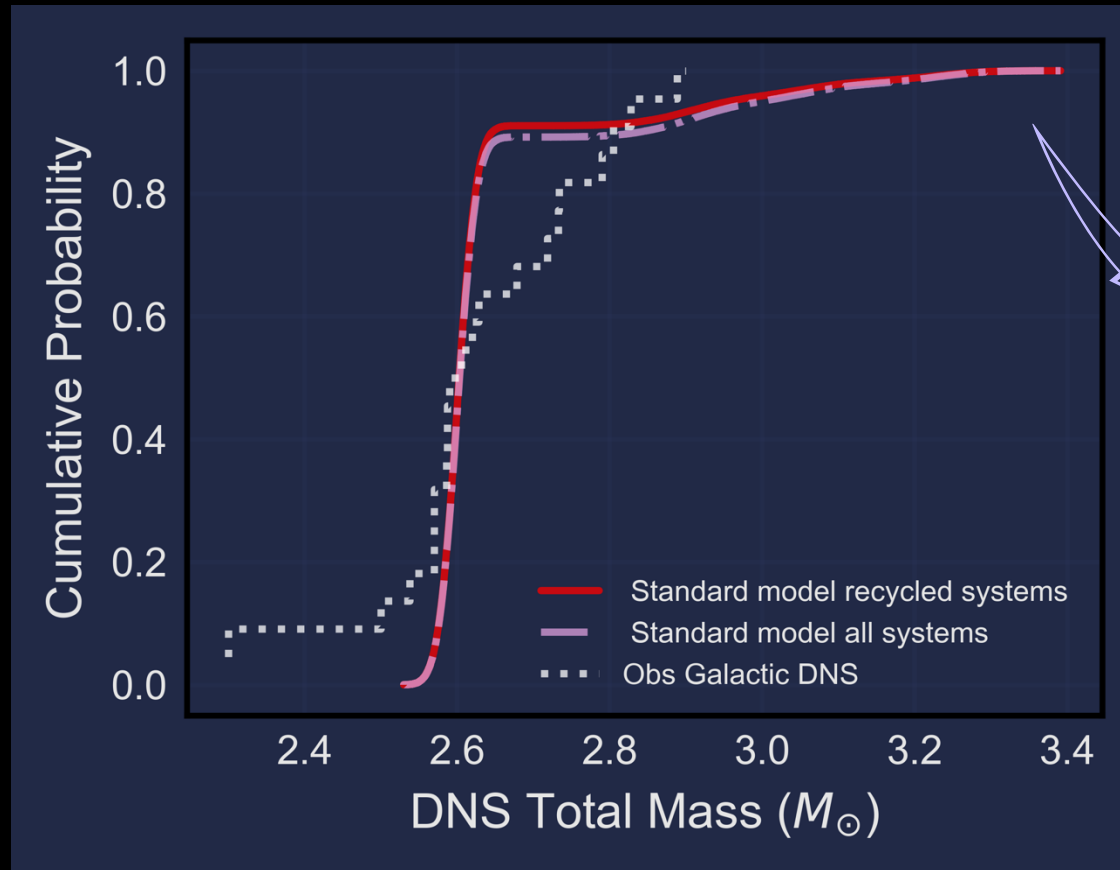
Kick velocity for the corresponding NS2 as a function of CO core mass

Combined P_{orb} -e Distribution of all models

Standard Model



DNS Total Mass Distribution Standard Model



Overproduce high mass systems
than observed

Assumed remnant mass and
CO core mass relation